RF Transmission, Reflections, and Standing Waves

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I. OBJECTIVE

(a) To determine the phase velocity and velocity factor of a signal in a coaxial cable from the frequency spacing between consecutive maxima or minima produced by reflection from an open-ended cable.

(b) To study signal reflection from an open-ended cable and from a cable terminated with a 50 Ω load.

(c) To estimate the magnitude of the reflection coefficient and verify that the coaxial cable is approximately matched to a 50 Ω termination, indicating that its characteristic impedance is close to 50Ω .

(d) To calculate the inductance and capacitance per unit length using the measured phase velocity and the approximate value $Z_0 \approx 50\Omega$.

(e) To measure the insertion loss and attenuation per unit length of the coaxial cable.

II. INTRODUCTION

Radio-frequency (RF) signals are electromagnetic waves used for the transmission of information and energy over electrical transmission lines. At sufficiently high frequencies, the physical dimensions of the connecting cable become significant compared with the wavelength of the signal, and the cable can no longer be treated as an ideal wire. Instead, the signal propagates along the cable as an electromagnetic wave, with a finite propagation velocity determined by the electrical and dielectric properties of the transmission line.

A coaxial cable is a widely used transmission line for RF applications because its geometry provides controlled electromagnetic propagation and shielding from external interference. The coaxial cable used in this experiment supports transverse electromagnetic (TEM) propagation, in which the electric and magnetic fields are transverse to the direction of propagation. The signal velocity in the cable is lower than the speed of light in vacuum because of the dielectric material between the conductors.

When an RF signal travelling along a transmission line reaches a load whose impedance differs from the characteristic impedance of the cable, part of the incident signal is reflected back toward the source. The incident and reflected waves interfere with each other, producing constructive and destructive interference. Consequently, the voltage amplitude measured at a fixed point varies between maxima and minima. In this experiment, the oscilloscope is connected at a Tee junction, so the spatial standing-wave pattern along the cable is not directly observed. Instead, the signal frequency is swept, causing the round-trip phase of the reflected wave to change and producing a periodic variation in the measured amplitude.

The periodic variation in amplitude provides a means of experimentally determining the propagation characteristics of the cable. In particular, the frequency spacing between consecutive maxima or minima is related to the round-trip propagation time of the reflected signal. From this frequency spacing and the measured physical length of the cable, the phase velocity and velocity factor of the RF signal can be calculated.

The experiment also investigates the reflection coefficient of the transmission line by comparing an open-ended cable with a cable terminated by a 50 Ω load. An open circuit produces strong reflection, whereas a termination matched to the characteristic impedance produces ideally no reflection. Therefore, a significant reduction in the amplitude variation when the 50 Ω termination is connected provides evidence that the characteristic impedance of the cable is approximately 50 Ω. The measured phase velocity and approximate characteristic impedance can further be used to determine the distributed inductance and capacitance of the cable. Finally, the attenuation introduced by the cable is determined by comparing the RF power measured through a short reference connection with the power measured after inserting the cable under test. This provides the insertion loss and attenuation per unit length of the cable. Thus, the experiment provides an experimental study of fundamental transmission-line phenomena, including RF signal propagation, impedance mismatch, signal reflection, interference and standing-wave formation, characteristic impedance, phase velocity, distributed inductance and capacitance, and cable attenuation. These measurements demonstrate how the observable frequency-dependent response of a practical coaxial cable can be related to its fundamental electrical and propagation parameters.

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III. THEORY

A. Transmission Lines and RF Signal Propagation

A transmission line is a physical structure used to guide electromagnetic energy from a source to a load. At radio frequencies, the physical dimensions of the connecting cable can become comparable to the wavelength of the electrical signal. Consequently, the cable can no longer be treated simply as an ideal conducting wire with negligible propagation delay. Instead, the signal must be considered as an electromagnetic wave propagating along the transmission line.

The coaxial cable used in this experiment supports transverse electromagnetic (TEM) propagation. In TEM propagation, both the electric field $\mathbf{E}$ *and the magnetic field* $\mathbf{B}$ *are transverse to the direction of propagation. If the cable axis is taken to be the z-direction, the electromagnetic energy propagates along the cable while the electric and magnetic fields remain perpendicular to this direction.*

The propagation velocity of the signal in a coaxial cable is generally lower than the speed of light in vacuum. This reduction occurs because the electromagnetic wave propagates through the dielectric material between the inner and outer conductors. The relevant velocity of propagation is therefore the phase velocity, denoted by v_p*.*

An important property of a transmission line is its characteristic impedance, Z_0*. The characteristic impedance describes the voltage-to-current ratio of a travelling wave propagating along an infinitely long transmission line, or equivalently a line terminated in its characteristic impedance. For the coaxial cable used in this experiment, the characteristic impedance is expected to be approximately* $50\,\Omega$*.*

B. Reflection of an RF Signal

When a travelling electromagnetic wave reaches the end of a transmission line, its behaviour depends on the impedance of the termination. Let the load impedance be Z_L *and the characteristic impedance of the cable be* Z_0*.*

If

$$Z_L = Z_0, \tag{1}$$

the cable is said to be matched. In this case, the incident wave is completely absorbed by the load and there is ideally no reflected wave.

However, when

$$Z_L \neq Z_0, \tag{2}$$

the load is mismatched to the transmission line. A portion of the incident wave is then reflected back toward the source. The voltage measured at any point on the cable is consequently the superposition of the incident and reflected waves.

Thus, the measured voltage can be represented schematically as

$$V_{\text{osc}} = V_{\text{incident}} + V_{\text{reflected}}. \tag{3}$$

The relative phase between the incident and reflected waves determines the measured amplitude. When the two waves arrive at the observation point in phase, constructive interference occurs and the measured amplitude becomes large. Conversely, when the waves are out of phase by 180°*, destructive interference occurs and the measured amplitude becomes small.*

This produces the characteristic maxima and minima associated with a standing-wave pattern.

It is important to note that in the present experiment the oscilloscope is connected to the Tee junction. Therefore, the oscilloscope does not directly display the spatial variation of voltage along the cable. Instead, the frequency of the signal is varied while observing the voltage at this fixed point. As the frequency changes, the phase accumulated by the reflected wave during its round trip changes. Consequently, the voltage amplitude measured at the Tee junction varies periodically with frequency.

C. Formation of Standing-Wave Maxima and Minima

Consider an incident wave and a reflected wave having amplitudes $|V_{\text{incident}}|$ *and* $|V_{\text{reflected}}|$*, respectively.*

When the two waves are in phase, their amplitudes add constructively. The maximum voltage amplitude is therefore

$$V_{\max} = |V_{\text{incident}}| + |V_{\text{reflected}}|. \tag{4}$$

When the two waves differ in phase by 180°*, they interfere destructively. Assuming that the incident-wave amplitude is greater than or equal to the reflected-wave amplitude, the minimum voltage amplitude is*

$$V_{\min} = |V_{\text{incident}}| - |V_{\text{reflected}}|. \tag{5}$$

The two equations can be added together:

$$V_{\max} + V_{\min} = 2|V_{\text{incident}}|. \tag{6}$$

Hence,

$$|V_{\text{incident}}| = \frac{V_{\max} + V_{\min}}{2}. \tag{7}$$

Similarly, subtracting the minimum amplitude from the maximum amplitude gives

$$V_{\max} - V_{\min} = 2|V_{\text{reflected}}|, \tag{8}$$

and therefore

$$|V_{\text{reflected}}| = \frac{V_{\max} - V_{\min}}{2}. \tag{9}$$

These relations allow the incident and reflected wave amplitudes to be estimated experimentally from the maximum and minimum voltages measured during a frequency sweep.

D. Phase Velocity Measurement

The phase velocity of the signal can be determined from the periodic variation in the measured amplitude with frequency.

Let the physical length of the coaxial cable be ℓ and let the phase velocity of the signal in the cable be v_p. A reflected wave travels from the Tee junction to the far end of the cable and then returns to the Tee junction. Therefore, the total distance travelled by the reflected wave during one round trip is

$$d_{\text{RT}} = 2\ell. \tag{10}$$

The corresponding round-trip propagation time is

$$\tau = \frac{2\ell}{v_p}. \tag{11}$$

The reflected wave accumulates a phase during this round trip. If the signal frequency is f, the round-trip phase is

$$\phi_{\text{RT}} = 2\pi f\tau + \phi_\Gamma, \tag{12}$$

where ϕ_Γ is the phase shift introduced by the termination.

Using

$$\tau = \frac{2\ell}{v_p}, \tag{13}$$

the round-trip phase can also be written as

$$\phi_{\text{RT}} = \frac{4\pi f\ell}{v_p} + \phi_\Gamma. \tag{14}$$

For an ideal open circuit, the reflection coefficient is positive and therefore the termination introduces no additional phase reversal. Hence,

$$\phi_\Gamma = 0 \tag{15}$$

for the ideal open-circuit case considered in the manual.

As the frequency is varied, the round-trip phase changes. A maximum in the measured amplitude occurs whenever the incident and reflected waves are in phase. The next maximum occurs when the round-trip phase has increased by 2π. The same condition applies to two consecutive minima.

If the frequency separation between two consecutive maxima or two consecutive minima is Δf, then

$$2\pi\Delta f\,\tau = 2\pi. \tag{16}$$

Cancelling 2π gives

$$\Delta f\,\tau = 1, \tag{17}$$

so that

$$\tau = \frac{1}{\Delta f}. \tag{18}$$

Substituting this result into

$$\tau = \frac{2\ell}{v_p}, \tag{19}$$

gives

$$\frac{2\ell}{v_p} = \frac{1}{\Delta f}. \tag{20}$$

Therefore, the phase velocity is

$$\boxed{v_p = 2\ell\Delta f}. \tag{21}$$

Thus, the phase velocity can be determined experimentally by measuring the physical cable length and the frequency spacing between two consecutive maxima or minima in the measured amplitude-frequency curve.

For improved accuracy, the frequency spacing can be determined over several consecutive maxima or minima and averaged.

E. Velocity Factor

The velocity factor (VF) expresses the propagation velocity in the cable relative to the speed of light in vacuum. It is defined as

$$\boxed{\text{VF} = \frac{v_p}{c}}. \tag{22}$$

where

$$c \approx 3 \times 10^8 \,\mathrm{m/s} \tag{23}$$

is the speed of light in vacuum.

Since the signal travels more slowly through the dielectric-filled coaxial cable than through vacuum,

$$v_p < c, \tag{24}$$

and therefore the velocity factor is less than unity.

The experimentally measured velocity factor therefore provides a convenient way of characterizing how rapidly an RF signal propagates through the cable compared with electromagnetic propagation in vacuum.

F. Reflection Coefficient

The amount of signal reflected from a load is quantified by the voltage reflection coefficient, Γ*. It is defined in terms of the load impedance* Z_L *and the characteristic impedance* Z_0 *as*

$$\boxed{\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}}. \tag{25}$$

The magnitude $|\Gamma|$ *describes the relative amplitude of the reflected voltage wave compared with the incident voltage wave.*

Three important termination cases are:

1. ***Open circuit:*** *For an open circuit,*

$$Z_L \to \infty, \tag{26}$$

and therefore

$$\Gamma = +1. \tag{27}$$

This corresponds to complete reflection without a phase reversal of the voltage wave.

2. ***Short circuit:*** *For a short circuit,*

$$Z_L = 0, \tag{28}$$

giving

$$\Gamma = -1. \tag{29}$$

The magnitude of the reflection coefficient is again unity, but the reflected voltage wave undergoes a phase reversal.

3. ***Matched termination:*** *If*

$$Z_L = Z_0, \tag{30}$$

then

$$\Gamma = 0. \tag{31}$$

In this ideal case there is no reflected wave.

In the present experiment, an open-circuit termination and a $50\,\Omega$ *termination are compared. The open circuit produces a strong reflected signal, whereas a* $50\,\Omega$ *termination is expected to substantially reduce the reflected signal if the characteristic impedance of the cable is close to* $50\,\Omega$*.*

G. Experimental Determination of the Reflection Coefficient

The magnitude of the reflection coefficient can be estimated directly from the measured maximum and minimum amplitudes.

From the previous relations,

$$|V_{\mathrm{incident}}| = \frac{V_{\max} + V_{\min}}{2} \tag{32}$$

and

$$|V_{\mathrm{reflected}}| = \frac{V_{\max} - V_{\min}}{2}. \tag{33}$$

The magnitude of the reflection coefficient is defined as

$$|\Gamma| = \frac{|V_{\mathrm{reflected}}|}{|V_{\mathrm{incident}}|}. \tag{34}$$

Substitution of the above expressions gives

$$|\Gamma| = \frac{(V_{\max} - V_{\min})/2}{(V_{\max} + V_{\min})/2}. \tag{35}$$

Therefore,

$$\boxed{|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}}. \tag{36}$$

For the open-ended cable, this quantity is expected to be relatively large because an open circuit produces strong reflection. For the $50\,\Omega$ *termination, the value should be substantially smaller if the cable is approximately matched to* $50\,\Omega$*.*

The experiment therefore does not directly determine Z_0 *from a general set of different loads. Instead, the manual specifies comparison of the open-circuit reflection with the reflection obtained using a* $50\,\Omega$ *termination. A substantial reduction in the reflected signal for the* $50\,\Omega$ *termination provides experimental evidence that the cable characteristic impedance is close to* $50\,\Omega$*.*

H. Characteristic Impedance of the Coaxial Cable

The characteristic impedance Z_0 is an intrinsic property of the transmission line. It determines the relationship between voltage and current for a travelling wave and also determines the termination required to eliminate reflections.

For this experiment, the approximate characteristic impedance is established experimentally by comparing the reflection obtained from an open circuit with that obtained from a $50\,\Omega$ load.

If

$$|\Gamma_{50}| \ll |\Gamma_{\text{open}}|, \tag{37}$$

then the $50\,\Omega$ termination produces much less reflection than the open termination. This indicates that the load is much better matched to the cable and hence that

$$\boxed{Z_0 \approx 50\,\Omega}. \tag{38}$$

This experimentally established approximation is subsequently used in calculating the distributed inductance and capacitance of the cable.

I. Inductance and Capacitance per Unit Length

A transmission line possesses distributed electrical properties. In particular, the coaxial cable has an inductance per unit length, denoted by L', and a capacitance per unit length, denoted by C'.

The units of these quantities are

$$L' : \text{H/m} \tag{39}$$

and

$$C' : \text{F/m}. \tag{40}$$

For a low-loss transmission line, the phase velocity is related to these distributed parameters by

$$v_p = \frac{1}{\sqrt{L'C'}}. \tag{41}$$

The characteristic impedance is related to the same parameters by

$$Z_0 = \sqrt{\frac{L'}{C'}}. \tag{42}$$

The two equations can be combined to determine L' and C' from the experimentally measured phase velocity and the approximate characteristic impedance.

Starting with

$$Z_0 = \sqrt{\frac{L'}{C'}}, \tag{43}$$

squaring both sides gives

$$Z_0^2 = \frac{L'}{C'}. \tag{44}$$

The phase-velocity relation gives

$$v_p = \frac{1}{\sqrt{L'C'}}, \tag{45}$$

or

$$v_p^2 L'C' = 1. \tag{46}$$

Combining these relations gives the expressions

$$\boxed{L' = \frac{Z_0}{v_p}} \tag{47}$$

and

$$\boxed{C' = \frac{1}{Z_0 v_p}}. \tag{48}$$

Thus, once v_p has been obtained from the frequency-spacing measurement and Z_0 has been approximated as $50\,\Omega$, both the inductance and capacitance per unit length can be calculated.

The values of L' and C' describe the distributed electrical behaviour of the coaxial cable and are fundamental parameters governing signal propagation along the transmission line.

J. Cable Attenuation and Insertion Loss

In addition to propagation delay and reflection, a practical coaxial cable introduces signal loss. As an RF signal propagates through the cable, its power decreases because of attenuation in the transmission line.

In this experiment, the signal power is measured using a spectrum analyser and is expressed in dBm.

To determine the loss introduced by the cable, a short reference connection is first used. The measured reference power is denoted by P_{ref}. The short reference connection is then replaced by the cable under test, and the resulting measured power is denoted by P_{cable}.

The insertion loss of the cable is defined as

$$\boxed{IL = P_{\text{ref}} - P_{\text{cable}}}. \tag{49}$$

Since both quantities are measured in dBm, their difference directly gives the insertion loss in dB.

For example, if the reference measurement gives a larger power than the measurement obtained after inserting the cable, then

$$P_{\text{ref}} > P_{\text{cable}}, \tag{50}$$

and consequently

$$IL > 0. \tag{51}$$

This positive value represents the reduction in signal power introduced by the cable and associated connection.

The attenuation per unit length is obtained by dividing the insertion loss by the physical cable length:

$$\boxed{\alpha = \frac{IL}{\ell}} \tag{52}$$

with units of dB/m.

Therefore, the measured attenuation depends on both the total insertion loss and the length of the cable under test. Measurements at several frequencies can additionally be used to investigate how the attenuation changes with frequency.

K. Overall Physical Interpretation

The experiment combines the effects of propagation, reflection, interference, and loss in a single RF transmission-line system.

When an RF signal is launched into the coaxial cable, it propagates toward the termination with phase velocity v_p. If the termination is mismatched to the characteristic impedance Z_0, part of the signal is reflected back toward the source. The incident and reflected waves then superpose at the Tee junction, producing constructive or destructive interference depending on their relative phase.

Because the round-trip phase depends on frequency, sweeping the signal frequency changes the relative phase between the incident and reflected waves. The oscilloscope therefore records a periodic variation in voltage amplitude as a function of frequency. The spacing between consecutive maxima or minima provides the round-trip delay and hence allows the phase velocity and velocity factor to be determined.

The relative magnitudes of the maxima and minima provide an estimate of the reflection coefficient. Comparing the reflection from an open circuit with that from a $50\,\Omega$ termination allows the cable to be tested for approximate $50\,\Omega$ matching. The measured phase velocity and characteristic impedance then provide the distributed inductance and capacitance per unit length. Finally, comparison of the reference and cable powers using a spectrum analyser gives the insertion loss and attenuation per unit length.

Thus, the experiment provides an experimental connection between the measurable frequency-dependent voltage response of a coaxial cable and its fundamental transmission-line parameters: propagation velocity, velocity factor, reflection coefficient, characteristic impedance, distributed inductance, distributed capacitance, and attenuation.

IV. COMPONENTS REQUIRED

- *RF signal generator*
- *Digital storage oscilloscope (DSO)*
- *Spectrum analyser*
- *RF Tee connector*
- *Coaxial cable under test*
- *Short reference coaxial cable*
- *50 Ω termination*
- *Open-circuit termination, obtained by leaving the cable end unconnected*
- *Suitable RF adapters and connectors*

V. EXPERIMENTAL SETUP

The experimental setup consists of an RF signal generator, a digital storage oscilloscope (DSO), a spectrum analyser, an RF Tee connector, the coaxial cable under test, a short reference coaxial cable, a $50\,\Omega$ termination, an open-circuit termination, and suitable RF adapters and connectors. The components specified in the laboratory manual are used to investigate signal propagation, reflection, standing-wave behaviour, and attenuation in the coaxial cable.

A. Setup for Reflection and Standing-Wave Measurements

For the phase-velocity and reflection measurements, the RF signal generator is connected to the coaxial cable under test through an RF Tee connector. The oscilloscope is connected to the third port of the Tee junction to measure the voltage amplitude of the signal at this fixed observation point. The far end of the coaxial cable is terminated according to the measurement being performed.

For the open-circuit measurement, the far end of the coaxial cable is left unconnected. The signal generator supplies a sinusoidal RF signal of constant amplitude, while its frequency is varied over the required range. The reflected signal from the open end travels back towards the Tee junction and interferes with the incident signal. As a

result, the voltage amplitude observed on the oscilloscope varies periodically with frequency.

The oscilloscope input impedance is set to $1\,\mathrm{M}\Omega$ *so that the measurement instrument does not significantly load the Tee junction and alter the observed reflection pattern. The signal-generator amplitude is kept constant throughout the frequency sweep.*

The arrangement can be represented schematically as

$$\textit{RF Signal Generator} \rightarrow \textit{Tee Junction} \rightarrow \textit{Coaxial Cable} \rightarrow \textit{Termination}, \tag{53}$$

with the oscilloscope connected to the Tee junction.

For the reflection-coefficient and approximate characteristic-impedance measurement, the same setup is retained. First, the far end of the cable is left open and the maximum and minimum voltage amplitudes are measured over one complete ripple period. The open termination is then replaced by a $50\,\Omega$ *termination, without changing the signal-generator amplitude or the frequency range. The corresponding maximum and minimum amplitudes are measured again. Comparison of the resulting reflection coefficients provides an experimental indication of whether the cable is approximately matched to a* $50\,\Omega$ *load.*

B. Setup for Cable Attenuation Measurement

For the attenuation measurement, the signal generator is connected to the spectrum analyser through a short reference coaxial connection and the required RF adapters. A suitable test frequency and input power are selected and kept constant during the measurement.

First, the power measured using the short reference connection is recorded as P_{ref} *in dBm. The short reference connection is then replaced by the coaxial cable under test, while keeping the remaining arrangement unchanged. The resulting measured power is recorded as* P_{cable} *in dBm.*

The insertion loss of the cable is subsequently determined from the difference between the reference and cable powers:

$$IL = P_{\mathrm{ref}} - P_{\mathrm{cable}}, \tag{54}$$

and the attenuation per unit length is calculated using

$$\alpha = \frac{IL}{\ell}, \tag{55}$$

where ℓ *is the physical length of the cable under test.*

VI. OBSERVATIONS

The experimental observations were recorded for the different stages of the RF transmission-line experiment. The measured quantities include the RMS voltage as a function of frequency, the maximum and minimum voltages for different termination conditions, and the reference and cable powers used to determine the insertion loss and attenuation of the coaxial cable.

A. Part A: Frequency Dependence of RMS Voltage

In Part A, the frequency of the RF signal was varied and the corresponding RMS voltage was measured using the oscilloscope. The coaxial cable used for this measurement had a measured length of

$$\ell = 305\ \mathrm{cm} = 3.05\ \mathrm{m}. \tag{56}$$

The measured RMS voltage as a function of frequency is presented in Tables I and II. The data have been divided into two tables to maintain a compact and readable format.

TABLE I: Measured RMS voltage as a function of frequency for Part A (first frequency range).

Frequency (MHz)	V_{RMS} (V)
10	1.210
11	1.060
12	0.881
13	0.720
14	0.550
15	0.385
16	0.219
17	0.084
17.2	0.076
17.5	0.091
18	0.156
19	0.309
20	0.468
21	0.626
22	0.766
23	0.907
24	1.050
25	1.160
26	1.300
27	1.400
28	1.470
29	1.500
30	1.630
31	1.680
32	1.730
33	1.770
34	1.780

The measured RMS voltage initially decreases with increasing frequency, falling from 1.210 V *at* 10 MHz *to*

TABLE II: Measured RMS voltage as a function of frequency for Part A (second frequency range).

Frequency (MHz)	V_{RMS} (V)
35	1.750
36	1.710
37	1.650
38	1.590
39	1.540
40	1.480
41	1.400
42	1.300
43	1.190
44	1.070
45	0.960
46	0.854
47	0.736
48	0.612
49	0.480
50	0.335
51	0.207
52	0.120
53	0.187
54	0.322
55	0.480
56	0.640
57	0.795
58	0.943
59	1.080
60	1.200

a minimum value of 0.076 V *at* 17.2 MHz. *The voltage subsequently increases and reaches a maximum value of* 1.780 V *at* 34 MHz. *It then decreases again and reaches a second minimum of* 0.120 V *at* 52 MHz, *after which the voltage increases with frequency.*

The experimentally observed extrema are summarized in Table III.

TABLE III: Observed extrema in the frequency-dependent RMS voltage.

Type	Frequency (MHz)	Voltage (V)
First minimum	17.2	0.076
Maximum	34	1.780
Second minimum	52	0.120

The periodic variation in the measured RMS voltage is associated with the interference between the incident and reflected waves in the transmission line. The frequencies corresponding to the successive maxima and minima are used in the subsequent analysis to determine the propagation characteristics of the cable.

B. Part B: Measurement of Maximum and Minimum Voltage

In Part B, the maximum and minimum voltages were measured for two different termination conditions: an open termination and a load termination. The same coaxial cable of length 305 cm *used in Part A was used for these measurements.*

The measured values are presented in Table IV.

TABLE IV: Measured minimum and maximum voltages for different termination conditions.

Termination	$V_{\min}$ (V)	$V_{\max}$ (V)
Open	0.0453	1.720
Load	0.933	1.060

For the open termination, the measured minimum and maximum voltages were 0.0453 V *and* 1.720 V, *respectively. For the load termination, the corresponding values were* 0.933 V *and* 1.060 V.

The measured maximum and minimum voltages are subsequently used to determine the magnitude of the voltage reflection coefficient using

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{57}$$

The calculated reflection coefficients are presented in the calculation and analysis section.

C. Part C: Distributed Transmission-Line Parameters

The distributed inductance per unit length, L', *and capacitance per unit length,* C', *were determined from the experimentally obtained transmission line parameters. The coaxial cable used for the measurements in the preceding sections had a length of* 305 cm.

The distributed parameters are calculated using

$$L' = \frac{Z_0}{v_p} \tag{58}$$

and

$$C' = \frac{1}{Z_0 v_p}, \tag{59}$$

where Z_0 *is the characteristic impedance and* v_p *is the phase velocity of the transmission line.*

The calculated values of L' *and* C' *are presented in the calculation section.*

D. Part D: Cable Loss Measurement

In Part D, the attenuation of the coaxial cable was measured by comparing the reference power with the power measured after inserting the cable under test. The measured length of the cable used for this experiment was

$$\ell = 91.5 \text{ cm} = 0.915 \text{ m}. \tag{60}$$

The reference power was first measured using the reference connection. The reference connection was then replaced by the 91.5 cm *cable under test, and the corresponding cable power was measured at different frequencies.*

The measured reference power and cable power are given in Table V.

TABLE V: Measured reference power and cable power for the cable under test.

Frequency (MHz)	P_{ref} (dBm)	P_{cable} (dBm)	IL (dB)
100	-1.40	-1.65	0.25
150	-1.40	-1.80	0.40
200	-1.40	-1.96	0.56
250	-1.40	-2.12	0.72
300	-1.40	-2.18	0.78

The insertion loss is calculated from the difference between the reference power and the power measured after inserting the cable:

$$IL = P_{\text{ref}} - P_{\text{cable}}. \tag{61}$$

Since both powers are expressed in dBm, their difference is expressed in dB.

The attenuation per unit length is calculated using

$$\alpha = \frac{IL}{\ell}, \tag{62}$$

where $\ell = 0.915$ m *is the measured cable length.*

The attenuation values obtained from the measurements are presented in Table VI.

TABLE VI: Measured attenuation of the cable at different frequencies.

Frequency (MHz)	α (dB/m)
100	0.2732
150	0.4372
200	0.6120
250	0.7869
300	0.8525

The measured insertion loss increases from 0.25 dB *at* 100 MHz *to* 0.78 dB *at* 300 MHz. *The corresponding attenuation increases from* 0.2732 dB/m *to* 0.8525 dB/m *over the same frequency range. Thus, the measured data show an overall increase in cable attenuation with increasing frequency.*

VII. CALCULATIONS

A. Calculation of Frequency Spacing

The phase velocity of the coaxial transmission line is determined from the frequency spacing between two successive minima in the standing-wave response.

From the measured standing-wave signal, the two successive minima occur at

$$f_1 = 17.2 \text{ MHz}, \tag{63}$$
$$f_2 = 52.0 \text{ MHz}. \tag{64}$$

Therefore, the frequency spacing is

$$\Delta f = f_2 - f_1 \tag{65}$$
$$= 52.0 - 17.2 \tag{66}$$
$$= 34.8 \text{ MHz}. \tag{67}$$

Thus,

$$\boxed{\Delta f = 34.8 \text{ MHz}}. \tag{68}$$

B. Calculation of Phase Velocity

The length of the coaxial cable used for the standing-wave measurement is

$$\ell = 305 \text{ cm} = 3.05 \text{ m}. \tag{69}$$

For two successive minima, the phase velocity is related to the frequency spacing by

$$v_p = 2\ell\Delta f. \tag{70}$$

Substituting the measured values,

$$v_p = 2(3.05)(34.8 \times 10^6) \tag{71}$$
$$= 2.123 \times 10^8 \text{ m/s}. \tag{72}$$

Therefore, the experimentally determined phase velocity is

$$\boxed{v_p = 2.12 \times 10^8 \text{ m/s}}. \tag{73}$$

C. Calculation of Velocity Factor

The velocity factor of the coaxial cable is defined as the ratio of the phase velocity in the cable to the speed of light in vacuum:

$$\mathrm{VF} = \frac{v_p}{c}, \tag{74}$$

where

$$c = 3.00 \times 10^8 \text{ m/s}. \tag{75}$$

Using the experimentally determined phase velocity,

$$\mathrm{VF} = \frac{2.123 \times 10^8}{3.00 \times 10^8} \tag{76}$$

$$= 0.7077. \tag{77}$$

Hence,

$$\boxed{\mathrm{VF} \approx 0.708}. \tag{78}$$

D. Calculation of Reflection Coefficient

The magnitude of the voltage reflection coefficient is determined from the maximum and minimum measured signal amplitudes using

$$|\Gamma| = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{79}$$

1. Open-Circuit Termination

For the open-circuit termination, the measured voltage extrema are

$$V_{\max,\mathrm{open}} = 1.720 \text{ V}, \tag{80}$$

$$V_{\min,\mathrm{open}} = 0.0453 \text{ V}. \tag{81}$$

Therefore,

$$|\Gamma_{\mathrm{open}}| = \frac{1.720 - 0.0453}{1.720 + 0.0453} \tag{82}$$

$$= \frac{1.6747}{1.7653} \tag{83}$$

$$= 0.9487. \tag{84}$$

Thus,

$$\boxed{|\Gamma_{\mathrm{open}}| \approx 0.949}. \tag{85}$$

2. 50 Ω Termination

For the 50 Ω termination, the measured voltage extrema are

$$V_{\max,50} = 1.060 \text{ V}, \tag{86}$$

$$V_{\min,50} = 0.933 \text{ V}. \tag{87}$$

Therefore,

$$|\Gamma_{50}| = \frac{1.060 - 0.933}{1.060 + 0.933} \tag{88}$$

$$= \frac{0.127}{1.993} \tag{89}$$

$$= 0.0637. \tag{90}$$

Hence,

$$\boxed{|\Gamma_{50}| \approx 0.0637}. \tag{91}$$

E. Determination of Characteristic Impedance

The reflection coefficients obtained for the two termination conditions are

$$|\Gamma_{\mathrm{open}}| \approx 0.949, \tag{92}$$

$$|\Gamma_{50}| \approx 0.0637. \tag{93}$$

The reflection coefficient for the 50 Ω termination is much smaller than that for the open-circuit termination:

$$|\Gamma_{50}| \ll |\Gamma_{\mathrm{open}}|. \tag{94}$$

This indicates that the 50 Ω termination provides a substantially better impedance match to the transmission line. Therefore, in accordance with the experimental procedure, the characteristic impedance of the cable is taken to be approximately

$$\boxed{Z_0 \approx 50 \ \Omega}. \tag{95}$$

The comparison of the two reflection coefficients therefore verifies that the coaxial cable behaves approximately as a 50 Ω transmission line.

F. Calculation of Inductance per Unit Length

For a lossless transmission line, the distributed inductance is related to the characteristic impedance and phase velocity by

$$L' = \frac{Z_0}{v_p}. \tag{96}$$

Using

$$Z_0 = 50\ \Omega, \tag{97}$$
$$v_p = 2.123 \times 10^8\ \text{m/s}, \tag{98}$$

we obtain

$$L' = \frac{50}{2.123 \times 10^8} \tag{99}$$
$$= 2.355 \times 10^{-7}\ \text{H/m}. \tag{100}$$

Therefore,

$$\boxed{L' \approx 2.36 \times 10^{-7}\ \text{H/m}} \tag{101}$$

or equivalently,

$$\boxed{L' \approx 236\ \text{nH/m}}. \tag{102}$$

G. Calculation of Capacitance per Unit Length

The distributed capacitance of the transmission line is calculated using

$$C' = \frac{1}{Z_0 v_p}. \tag{103}$$

Substituting the measured phase velocity and the approximate characteristic impedance,

$$C' = \frac{1}{(50)(2.123 \times 10^8)} \tag{104}$$
$$= 9.42 \times 10^{-11}\ \text{F/m}. \tag{105}$$

Therefore,

$$\boxed{C' \approx 9.42 \times 10^{-11}\ \text{F/m}} \tag{106}$$

or

$$\boxed{C' \approx 94.2\ \text{pF/m}}. \tag{107}$$

H. Calculation of Insertion Loss

For the attenuation measurement, a coaxial cable of length

$$\ell = 91.5\ \text{cm} = 0.915\ \text{m} \tag{108}$$

was used.

The insertion loss is calculated from the difference between the reference power and the power measured with the cable connected:

$$IL = P_{\text{ref}} - P_{\text{cable}}. \tag{109}$$

1. At 100 MHz

The measured powers are

$$P_{\text{ref}} = -1.40\ \text{dBm}, \tag{110}$$
$$P_{\text{cable}} = -1.65\ \text{dBm}. \tag{111}$$

Therefore,

$$IL = (-1.40) - (-1.65) \tag{112}$$
$$= 0.25\ \text{dB}. \tag{113}$$

The attenuation per unit length is

$$\alpha = \frac{IL}{\ell} \tag{114}$$
$$= \frac{0.25}{0.915} \tag{115}$$
$$= 0.273\ \text{dB/m}. \tag{116}$$

Thus,

$$\boxed{\alpha_{100} = 0.273\ \text{dB/m}}. \tag{117}$$

2. At 150 MHz

The measured cable power is

$$P_{\text{cable}} = -1.80\ \text{dBm}. \tag{118}$$

Hence,

$$IL = (-1.40) - (-1.80) \tag{119}$$
$$= 0.40\ \text{dB}. \tag{120}$$

Therefore,

$$\alpha = \frac{0.40}{0.915} \tag{121}$$
$$= 0.437 \text{ dB/m}. \tag{122}$$

Thus,

$$\boxed{\alpha_{150} = 0.437 \text{ dB/m}}. \tag{123}$$

3. *At* 200 MHz

The measured cable power is

$$P_{\text{cable}} = -1.96 \text{ dBm}. \tag{124}$$

Therefore,

$$IL = (-1.40) - (-1.96) \tag{125}$$
$$= 0.56 \text{ dB}. \tag{126}$$

The attenuation is

$$\alpha = \frac{0.56}{0.915} \tag{127}$$
$$= 0.612 \text{ dB/m}. \tag{128}$$

Hence,

$$\boxed{\alpha_{200} = 0.612 \text{ dB/m}}. \tag{129}$$

4. *At* 250 MHz

The measured cable power is

$$P_{\text{cable}} = -2.12 \text{ dBm}. \tag{130}$$

Thus,

$$IL = (-1.40) - (-2.12) \tag{131}$$
$$= 0.72 \text{ dB}. \tag{132}$$

The attenuation per unit length is

$$\alpha = \frac{0.72}{0.915} \tag{133}$$
$$= 0.787 \text{ dB/m}. \tag{134}$$

Therefore,

$$\boxed{\alpha_{250} = 0.787 \text{ dB/m}}. \tag{135}$$

5. *At* 300 MHz

The measured cable power is

$$P_{\text{cable}} = -2.18 \text{ dBm}. \tag{136}$$

Therefore,

$$IL = (-1.40) - (-2.18) \tag{137}$$
$$= 0.78 \text{ dB}. \tag{138}$$

The attenuation per unit length is

$$\alpha = \frac{0.78}{0.915} \tag{139}$$
$$= 0.852 \text{ dB/m}. \tag{140}$$

Thus,

$$\boxed{\alpha_{300} = 0.852 \text{ dB/m}}. \tag{141}$$

VIII. GRAPHS

The experimental data were plotted to investigate the frequency-dependent behaviour of the transmission line. Three graphs were prepared from the measured and calculated quantities: the standing-wave signal amplitude as a function of frequency, the insertion loss as a function of frequency, and the attenuation per unit length as a function of frequency.

A. Standing-Wave Signal Amplitude versus Frequency

Figure 1 shows the variation of the standing-wave signal amplitude with frequency for the 305 cm *coaxial cable. The measured RMS voltage exhibits a periodic variation as the frequency is changed. This behaviour is characteristic of the interference between the incident and reflected waves travelling along the transmission line.*

The first minimum in the measured voltage occurs at approximately 17.2 MHz*, where the RMS voltage is* 0.076 V*. The voltage subsequently increases and reaches a maximum of* 1.780 V *at* 34 MHz*. It then decreases again, reaching a second minimum of* 0.120 V *at* 52 MHz*. The successive minima provide the frequency separation required for determining the phase velocity of the transmission line.*

The periodic maxima and minima observed in Figure 1 indicate the formation of a standing-wave pattern. The spacing between successive minima is related to the propagation velocity of the signal in the cable and is therefore used in the subsequent calculation of the phase velocity.

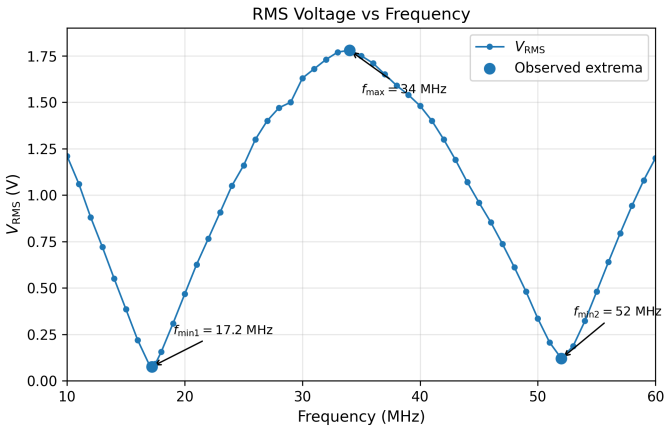


FIG. 1: Variation of standing-wave signal amplitude with frequency due to interference between incident and reflected waves for the 305 cm coaxial cable.

B. Insertion Loss versus Frequency

Figure 2 shows the variation of insertion loss with frequency for the 91.5 cm *coaxial cable. The insertion loss was determined from the difference between the measured reference power and the power measured after inserting the cable under test.*

The insertion loss increases progressively with frequency. At 100 MHz*, the measured insertion loss is* 0.25 dB*, whereas at* 300 MHz *it increases to* 0.78 dB*. The increase in insertion loss indicates that the cable introduces greater power loss at higher frequencies over the measured frequency range.*

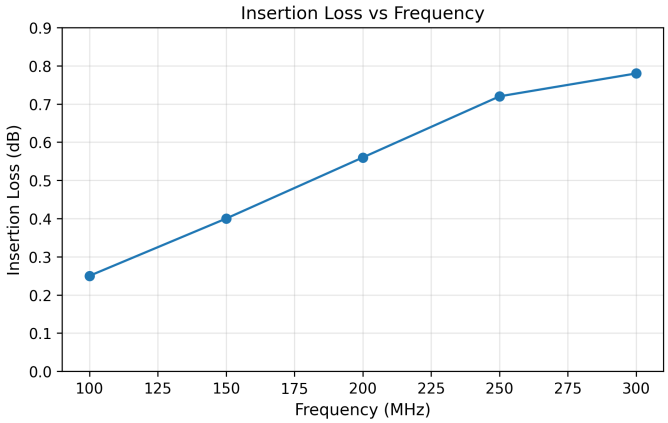


FIG. 2: Variation of insertion loss with frequency for the 91.5 cm coaxial cable.

The approximately increasing trend in Figure 2 shows that the transmission loss of the cable is frequency dependent. The measured insertion loss values are subsequently used to determine the attenuation per unit length of the cable.

C. Attenuation versus Frequency

Figure 3 shows the variation of attenuation per unit length with frequency for the 91.5 cm *coaxial cable. The attenuation was calculated by dividing the measured insertion loss by the length of the cable.*

The attenuation increases from approximately 0.273 dB/m *at* 100 MHz *to* 0.852 dB/m *at* 300 MHz*. Thus, the experimental results show an overall increase in attenuation with increasing frequency.*

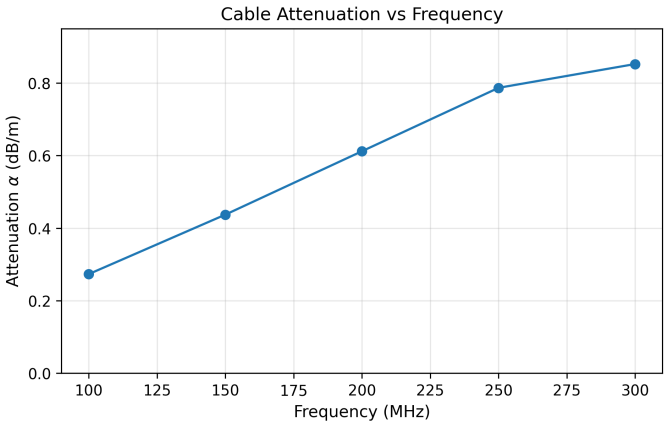


FIG. 3: Variation of attenuation per unit length with frequency for the 91.5 cm coaxial cable.

The increasing attenuation indicates that the cable exhibits greater transmission loss at higher frequencies. The attenuation plot therefore provides a direct representation of the frequency-dependent loss characteristics of the coaxial cable.

D. Overall Graphical Analysis

The three graphs together demonstrate the principal experimental characteristics of the transmission line. The standing-wave plot shows the periodic variation in signal amplitude produced by interference between incident and reflected waves. The locations of the voltage minima and maximum provide the frequency information required for determining the propagation characteristics of the cable.

The insertion-loss and attenuation plots show a consistent increase in transmission loss with frequency. In particular, the attenuation increases from 0.273 dB/m *at* 100 MHz *to* 0.852 dB/m *at* 300 MHz*. These graphical observations are consistent with the calculated transmission-line parameters and are used in the subsequent results and discussion.*

IX. RESULTS

A. Summary of Calculated Results

The principal calculated transmission-line parameters are summarized below.

TABLE VII: Summary of calculated transmission-line parameters.

Parameter	Experimental Result
Frequency spacing, Δf	34.8 MHz
Phase velocity, v_p	2.12×10^8 m/s
Velocity factor	0.708
$\lvert\Gamma_{\text{open}}\rvert$	0.949
$\lvert\Gamma_{50}\rvert$	0.0637
Characteristic impedance, Z_0	$\approx 50\ \Omega$
Inductance per unit length, L'	236 nH/m
Capacitance per unit length, C'	94.2 pF/m

The attenuation results for the 91.5 cm *cable are summarized in Table VIII.*

TABLE VIII: Calculated insertion loss and attenuation of the 91.5 cm coaxial cable.

Frequency (MHz)	P_{ref} (dBm)	P_{cable} (dBm)	α (dB/m)
100	−1.40	−1.65	0.273
150	−1.40	−1.80	0.437
200	−1.40	−1.96	0.612
250	−1.40	−2.12	0.787
300	−1.40	−2.18	0.852

The experimental results demonstrate the standing-wave, reflection, and attenuation characteristics of the coaxial transmission line.

Overall, the results verify the expected standing-wave behaviour and show that the transmission line is approximately a 50 Ω *cable with frequency-dependent attenuation.*

X. CONCLUSION

The experiment successfully demonstrated the fundamental characteristics of a coaxial transmission line, including standing-wave formation, signal propagation, impedance matching, distributed parameters, and frequency- dependent attenuation.

- *The plot of standing-wave signal amplitude versus frequency showed a periodic variation in the measured RMS voltage. The observed maxima and minima arise from interference between the incident and reflected waves on the transmission line. Two consecutive minima were observed at* 17.2 MHz *and* 52.0 MHz*, giving a frequency spacing of*

$$\Delta f = 34.8 \text{ MHz}. \tag{142}$$

This periodic behaviour confirms the formation of standing waves on the transmission line.

- *For the* 305 cm *cable, the measured phase velocity was*

$$v_p \approx 2.12 \times 10^8 \text{ m/s}, \tag{143}$$

corresponding to a velocity factor of

$$\text{VF} \approx 0.708. \tag{144}$$

Thus, the RF signal propagates through the cable at approximately 70.8% *of the speed of light in vacuum.*

- *The reflection coefficient for the open-circuit termination was approximately*

$$|\Gamma_{\text{open}}| \approx 0.949, \tag{145}$$

whereas for the 50 Ω *termination it was*

$$|\Gamma_{50}| \approx 0.0637. \tag{146}$$

The much smaller reflection obtained with the 50 Ω *termination verifies that the cable is approximately matched to a* 50 Ω *load. Therefore, its characteristic impedance can be taken as approximately*

$$Z_0 \approx 50\ \Omega. \tag{147}$$

This demonstrates the importance of impedance matching in minimizing reflections and maximizing power transfer.

- *Using the measured phase velocity and the approximate characteristic impedance, the distributed parameters of the cable were calculated as*

$$L' \approx 236 \text{ nH/m} \tag{148}$$

and

$$C' \approx 94.2 \text{ pF/m}. \tag{149}$$

These parameters represent the distributed inductive and capacitive properties of the coaxial transmission line and determine its propagation characteristics.

- *The attenuation measurements using the* 91.5 cm *cable showed that the attenuation increases with frequency. It increased from*

$$0.273 \text{ dB/m} \tag{150}$$

at 100 MHz *to*

$$0.852 \text{ dB/m} \tag{151}$$

at 300 MHz*. This indicates that the cable becomes increasingly lossy as the operating frequency increases.*

- *The main sources of measurement uncertainty include the limited resolution and accuracy of the oscilloscope when determining* $V_{\max}$ *and* $V_{\min}$*, uncertainty in locating the exact frequencies of the standing-wave minima, uncertainty in the physical measurement of cable length, connector and adapter losses, and imperfections in the* 50 Ω *termination. For the attenuation measurement, additional uncertainty may arise from the power measurement resolution and from differences between the reference and cable-connected configurations.*

XI. REFERENCES

1. *Lab Mannual of SPA617*
2. *Wikipedia*
3. *Datasheet of RG174 -Datasheet*
4. *AI tool for more conceptual understanding.*

XII. DECLARATION OF AI USAGE

I have used AI tools in the following ways for the completion of my lab report:

- *Formatting the theory content into various sub sections and concise writing of the same after I studied them from various sources. It helped me compile them well.*
- *Formatting of graph in python (lines along with labels) and the algorithm to find the cut off frequency from the graph.*

However ,the calculations, error analysis, data acquisition and processing, conclusions, key takeaways, results were all self written from scratch and there is absolutely no AI aid taken over those sections.

XIII. ACKNOWLEDGMENT

I would like to thank our course instructor Dr. Mugundhan V. , our REO - Jayasandhya Meenakshinathan and our TA - Banhiman Ghosh, without their help this lab experiment would have been very difficult to complete.